

CRYSS Input Parser Layer

Technical Documentation

CRYSS Project

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1 Overview

The CRYSS input parser layer is responsible for transforming raw slice-level experimental data into a structured, quality-controlled representation suitable for downstream optimisation, PCA analysis, and mechanistic modelling. This document describes the full processing pipeline, including replicate filtering, slice-level statistics, mixture-level aggregation, and final Excel output.

2 Data Model

Each experimental file contains a set of mixtures. Each mixture contains one or more slices, and each slice contains:

- Solid-phase composition X_s
- Liquor-phase composition X_l
- Mixture composition X_c
- Solid fraction R
- Slice identifier (e.g. EXP001, EXP002F5)

All compositions satisfy:

$$\sum_i X_s^{(i)} = 1, \quad \sum_i X_l^{(i)} = 1, \quad \sum_i X_c^{(i)} = 1.$$

3 Quality Control Pipeline

The parser applies a two-stage QC process:

3.1 Stage 1: Xc Filtering

Each slice is compared against the median mixture composition:

$$\Delta X_c^{(i)} = \left| X_{c,\text{slice}}^{(i)} - X_{c,\text{median}}^{(i)} \right|.$$

A slice is rejected if:

$$\max_i \Delta X_c^{(i)} > 0.05.$$

This removes gross composition errors.

3.2 Stage 2: Replicate Filtering

Replicates are compared against canonical mixture values using stricter analytical tolerances:

$$\max_i \left| X_{c,\text{slice}}^{(i)} - X_{c,\text{canonical}}^{(i)} \right| < 0.02,$$

$$|R_{\text{slice}} - R_{\text{canonical}}| < 0.10.$$

Slices failing either condition are rejected.

4 Slice-Level Statistics

For each accepted slice, the parser computes:

- Solid enrichment index:

$$SE = \sum_i (X_s^{(i)} - X_c^{(i)})^2$$

- Liquor enrichment index:

$$LE = \sum_i (X_l^{(i)} - X_c^{(i)})^2$$

- Solid-liquor separation:

$$SLS = \sum_i (X_s^{(i)} - X_l^{(i)})^2$$

- Slice purity index:

$$PI = \max_i X_s^{(i)}$$

These metrics quantify slice geometry, purity, and separation strength.

5 Mixture-Level Statistics (Stage 2)

Mixture-level aggregation is performed across all accepted slices.

5.1 Canonical Mixture Values

Canonical mixture values are defined as medians across slices:

$$X_{s,\text{mix}}^{(i)} = \text{median}_{\text{slices}}(X_s^{(i)}),$$

$$X_{l,\text{mix}}^{(i)} = \text{median}_{\text{slices}}(X_l^{(i)}),$$

$$R_{\text{mix}} = \text{median}_{\text{slices}}(R).$$

5.2 Slice-to-Slice Variation

Standard deviations quantify mixture consistency:

$$\sigma_s^{(i)} = \text{std}(X_s^{(i)}), \quad \sigma_l^{(i)} = \text{std}(X_l^{(i)}), \quad \sigma_R = \text{std}(R).$$

5.3 Lever-Rule Consistency

Lever-rule consistency is evaluated using:

$$X_{c,\text{mix}}^{(i)} = X_{s,\text{mix}}^{(i)} + X_{l,\text{mix}}^{(i)},$$

with median and standard deviation recorded.

5.4 Purity Index

A mixture-level purity index is defined as:

$$PI_{\text{mix}} = \text{mean}_{\text{slices}}(|X_s - X_l|).$$

6 Excel Output Structure

The parser produces a multi-sheet Excel workbook:

1. **slice_summary**: canonical slice values
2. **mixture_stats**: mixture-level medians, variation, purity
3. **slice_stats**: slice-level geometry metrics
4. **component_stats**: per-component trends (Stage 3)
5. **file_summary**: global metadata and QC totals

This structure mirrors the processing pipeline:

$$\text{slice} \rightarrow \text{mixture} \rightarrow \text{component} \rightarrow \text{file}.$$

7 Conclusion

This document provides a complete description of the CRYSS input parser layer. It defines the QC logic, slice-level metrics, mixture-level aggregation, and final output structure. As the GUI layer matures, this parser will operate behind the scenes, but its behaviour is fully documented here for future reference and reproducibility.